COUNTRY PROFILES

Nuclear Power in Switzerland

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- Switzerland has four nuclear reactors generating up to 40% of its electricity. Two large new units were planned.
- In June 2011 parliament resolved not to replace any reactors, and hence to phase out nuclear power gradually, and this was confirmed in a 2017 referendum.
- In August 2024, the Federal Council announced it would seek to lift the construction ban, citing climate

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targets, rising electricity demand, and geopolitical uncertainties.

4

OPERABLE REACTORS

2,973 MWe

0

REACTORS UNDER CONSTRUCTION

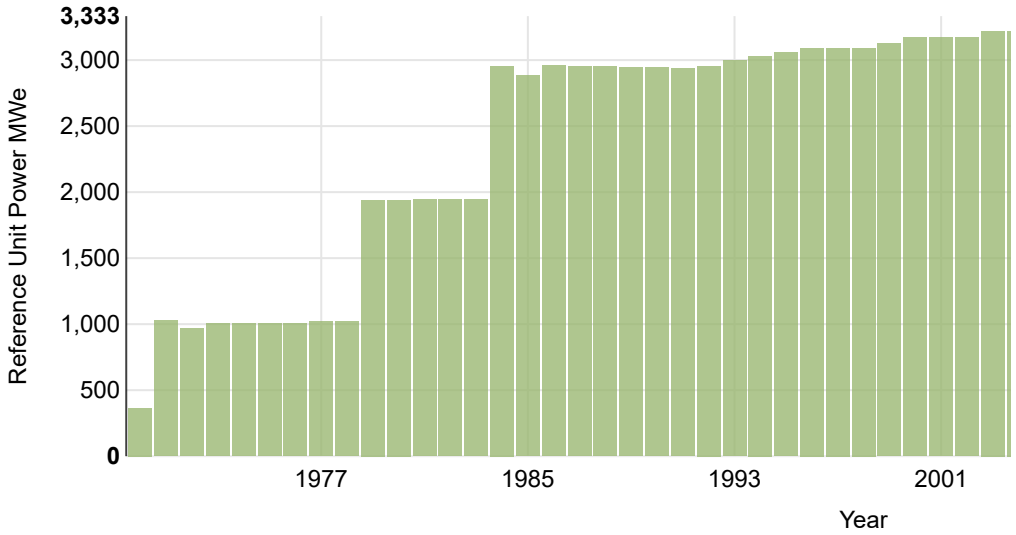
0 MWe

2

REACTORS SHUTDOWN

381 MWe

Operable nuclear power capacity



Electricity sector

Electricity consumption in Switzerland has been stable at 55-60 TWh for the last 10 years. Total exports and imports are largely balanced.

Total generation (in 2023): 73.9 TWh

Generation mix: hydro 41.2 TWh (56%); nuclear 24.4 TWh (33%); biofuels & waste (5.3 TWh (7%); solar 4.6 TWh (6%); natural gas 0.3 TWh; wind 0.2 TWh.

Import/export balance: 6.4 TWh net export (33.9 TWh exports, 22.5 TWh imports)

Total consumption: c. 56.1 TWh

Per capita consumption: c. 6300 kWh

Source: International Energy Agency and The World Bank. Data for year 2023.

Electricity consumption in Switzerland has been stable at 55-60 TWh for the last 10 years. Electricity is imported primarily from Germany, France and Austria and exported primarily to Italy. Total exports and imports are largely balanced. To ensure security of supply during winter months, Swiss utilities have long-term contracts with EDF to import 2500 MWe of French nuclear power, at a price premium. In March 2017 the National Council voted 136 to 52 to maintain this arrangement, but EDF terminated one such arrangement in June 2020 after Alpiq revoked its contractual arrangements citing force majeure during the height of the Covid-19 pandemic.

Nuclear power plants

Switzerland has three commercial nuclear power plants with a total of four operable reactors. Both Beznau and Gösgen produce district heating in addition to power. Beznau makes available 80 MW of heat to industry and homes over a 130 km network serving 11 towns – potentially 2.5 PJ/yr.

All operational Swiss reactors have had power uprates – the Beznau units from 350 MWe to 365 MWe, Gösgen from 920 MWe to 1010 MWe to 2014 (with major equipment replacement in 2013), and Leibstadt from 990 MWe progressively to 1220 MWe by 2013.

All operational Swiss reactors have unlimited-duration operating licences, as long as they can demonstrate safety to the regulator. Mühleberg, shutdown in December 2019, required a new licence every ten years, granted on condition that it met nuclear safety requirements. A 10-year licence was granted in December 2009; however, in 2012 the Federal Administrative Court ruled that the reactor should close in mid-2013 on safety grounds. It said that the issue of safety was too important to be the sole responsibility of the regulator, the Swiss Federal Nuclear Safety Inspectorate (ENSI), and that if BKW wanted to operate beyond mid-2013 it would need to make a strong case for that. BKW, along with the Energy Ministry, appealed, and in March 2013 the Supreme Court reversed the ruling.

BKW considered its options regarding safety-related upgrades to Mühleberg, and decided in October 2013 to spend some CHF 200

million on safety upgrades and close the reactor in 2019. Prolonging its operational lifetime to 2022 as initially planned would have involved higher costs in the context of "uncertainty surrounding political and regulatory trends". In May 2014 citizens of Bern canton effectively approved the 2019 closure date with a 63% vote rejecting an early shutdown proposal. This was the first Swiss public vote on nuclear power since the government 2011 decision to phase it out, and is consistent with a 2013 poll (see below). In January 2015 ENSI approved BKW's revised maintenance plan for Mühleberg, allowing the company to operate it until 2019.

Beznau 1&2

In 2015 concerns were raised about the manufacturing quality of the reactor pressure vessels of the Beznau reactors, and restart of unit 1 was postponed from July as investigations by ENSI continued. Unit 2 was checked and restarted in December 2015. Axpo said the testing was similar to that used on two Belgian reactors though the metallurgical irregularities were different, and the vessels had been made by Creusot Forge in France. Axpo invested CHF 700 million in unit 1 during its outage, and the same for unit 2, giving them prospective 60-year operating lives. Axpo submitted the safety case for unit 1 to ENSI in November 2016, providing the results of all the tests and analyses it had carried out over the past six months on the vessel in order to produce proof of physical integrity. Axpo said that the results confirmed the safety of the reactor for operation to 2030, but approval for restart was not granted until March 2018. In December 2024, Axpo announced a CHF 350 million investment enabling unit 2 to operate until 2032 and unit 1 until 2033.

Nuclear industry development

The country's first research reactor – the 10 MW SAPHIR – started up in 1957, having been bought from the USA, and it ran until 1993. A second unit – DIORIT (30 MW) – was designed and constructed indigenously and started up in 1960, running until 1977. In 1960 the Swiss government took over the research centre operating both reactors and in 1988 this became the Paul Scherrer Institute – a flagship research centre.

Construction of an experimental power reactor commenced in 1962 at Lucens. This was a 30 MWt, 7 MWe heavy-water moderated gas-cooled unit located in an underground cavern. It

started up in 1966 but experienced a core melt in 1969 and was written off.

In the 1960s it was evident that Swiss power demand was likely to exceed the potential for supply from hydro sources, so utilities proposed building coal- and oil-fired plants. However, this was strenuously opposed by environmental groups and others on the basis of compromising the hitherto clean power generation, so the government encouraged the utilities to plan for nuclear power.

The country's first commercial units were Beznau 1 – a Westinghouse pressurized water reactor ordered by NOK (Nordostschweizerische Kraftwerke AG) and soon duplicated (Beznau 2), and Mühleberg – a General Electric boiling water reactor ordered by BKW (Bernische Kraftwerke AG), located 15 km from Bern.

Following these three units, a consortium of utilities – Kernkraftwerk Gösgen (KKG) – ordered a large PWR from Siemens KWU for Gösgen and the same year another utility consortium (KKL) ordered a similar-sized General Electric BWR for Leibstadt.

A further unit (950 MWe) was proposed for Kaiseraugst near Basel, but this was abandoned in 1988 following anti-nuclear opposition arising from the Chernobyl accident, as was the Graben proposal (1140 MWe). Kaiseraugst investors were compensated CHF 350 million.

Research reactors

The Paul Scherrer Institute submitted an application in April 2013 to the Federal Office of Energy (FOE) for permission to decommission its zero-power Proteus research reactor, which was commissioned in 1968 and shut down in 2011. ENSI approved this in 2016. There is another very small research reactor operating at École Polytechnique Fédérale de Lausanne, and a third, pool-type, research reactor at the University of Basel, which is undergoing dismantling.

Energy policy

1990 to 2011

A ten-year moratorium on new plant construction was supported by 54.6% of the electorate during a national referendum in 1990.

Then in a unique 2003 referendum which would have been binding and written into the constitution, Swiss voters firmly rejected two anti-nuclear proposals which were originally put forward in 1998. "Electricity without Nuclear" was overtly to phase out nuclear power by 2014, while "Moratorium Plus" would have led to a similar outcome by, amongst other things, removing incentives to invest in and upgrade nuclear plants. Two-thirds of voters rejected the first proposal and 58% rejected the second, with practically all cantons refusing both.

In 2006 ATEL was looking for partners to build a further large nuclear power plant using proven technology and probably at an existing nuclear plant site. Also Axpo Holding AG had been studying sites for a new nuclear power plant – possibly Beznau.

The Swiss government announced early in 2007 that the existing five nuclear power reactors should be replaced in due course with new units. The new energy policy included renewables, energy efficiency and gas-fired plants, but had nuclear continuing to carry the main load apart from hydro, which is not amenable to expansion.

In October 2007 the Solothurn canton parliament called for “rapid construction of a nuclear power station in Niederamt” near Gösgen. In 2008, framework approval applications were filed for two projects with large new reactors.* Rationalising these, in December 2010 Axpo, Alpiq and BKW joined forces to form a joint planning company on an equal basis to pursue construction of two identical new nuclear power reactors of up to about 1600 MWe at two of the Niederamt, Beznau or Mühleberg sites. Axpo and one of its subsidiaries would take 59% of the power from the new reactors, Alpiq would take 25.5%, and BKW would have 15.5% – the ratio being according to the three partners' existing shares of Switzerland's nuclear power capacity.

* In 2007 the Resun joint venture was formed by NOK Axpo (57.75%), BKW FMB Energie (31.25%) and Centralschweizerische Kraftwerk (11%) to apply to construct two reactors of up to 1600 MWe each at Beznau and Mühleberg sites. ATEL (now Alpiq) was invited to join the venture, but did not do so at that stage. Then in December 2008 the Axpo Group and BKW FMB Energie filed framework permit applications for new nuclear units at Beznau and Mühleberg, a first step towards the replacement of the three small nuclear reactors there which were due to close 2019-22. Two identical new 1100 to

1700 MWe reactors were envisaged, with hybrid cooling systems to minimize water use. Bids for these were sought by April 2011.

Meanwhile, in June 2008 ATEL (now Alpiq) subsidiary Nuclear Power Plant Niederamt Ltd applied to the Federal Office of Energy for framework approval to build a new nuclear power plant in Niederamt near Gösgen. An advanced 1100 to 1600 MWe reactor was envisaged, with hybrid cooling system which would minimize water use. Estimated cost was €3.7 to 4.5 billion to be shared by partners, with start-up expected after 2020. It sounded out possible partners, including Axpo and BKW FMB Energie. There was strong local support for the project.

In November 2010 the Federal Nuclear Safety Inspectorate (ENSI) drew up definitive appraisals for the Swiss Federal Office of Energy saying that the Niederamt, Beznau and Mühleberg sites were suitable for the purpose of building new reactors. ENSI's findings on the applications were reviewed as part of a public enquiry in mid-2011. A federal decision on granting general authorizations for the plants had been expected but was postponed. The Federal Office of Energy was reported to favour construction of two new reactors, not three.

2011 onwards

In May 2011, following the Fukushima Daiichi accident in Japan, the Swiss federal government declared that the country's nuclear power plants would not be replaced. The National Council on 7 June 2011 voted 101 to 54 to endorse this, so that nuclear energy would be phased out in due course, with the last unit going offline by around 2034 (on the basis of a 50-year operating lifetime for the newest unit). The proposal was also approved by the upper house, the 46-member Council of States, by 3:1, though subject to ongoing review of technology options which might allow new plants.

Following the election in December 2011, the government was to produce a new energy policy without nuclear power and submit that to parliament. In 2014 the Energy Strategy 2050 was considered and the National Council energy committee called for the introduction of a system requiring operators to submit plans for improving the safety of reactors after 40 years of service. Subject to approval by the ENSI, this would enable reactors to continue in operation for a further 10-year period, with no limit to the number of 10-year extensions. If a reactor was judged unfit to continue in operation, the operator would receive no compensation. A Green Party initiative to limit the life of nuclear power stations to 45 years

was rejected. The policy promoted expansion of renewable energy including hydro, with subsidies, and proposed building some gas-fired plants – both CHP and CCGT – with subsidies, and raising electricity tariffs. The National Council largely endorsed the nuclear proposals in December 2014, so that after 40 years the nuclear power plant licence could be renewed for ten years and then possibly another ten years. The 'no new nuclear plants' policy was accepted by 115 to 77.

The Council of States (cantons) voted on the matter in September 2015, and agreed to avoid putting legal limits on the operating lives of reactors. It also rejected the National Council proposal supported by ENSI of requiring operators to submit a long-term operating concept every 10 years once a reactor reaches 40 years of service. The upper house also voted to impose a time limit on the federal renewable energy feed-in tariff subsidy scheme (KEV), and for switching some of the funds supporting wind and solar power to subsidize existing hydropower stations.

Following elections the previous month, in January 2016 the Energy Committee of the National Council voted against imposing any limits on reactor operating life, and against the requirement for 10-year plans and justification as proposed by ENSI, thus lining up with the Council of States. It also voted to reverse the ban on building new reactors, and voted against any lifetime limit for Beznau.

In November 2016 a referendum brought by the Green party proposed that nuclear plants be closed after a maximum 45 years in operation. This would have meant three of the five reactors closing in 2017 and the other two in 2024 and 2029. It failed by about 54:46, with voters expressing confidence in both operators and the safety authority, despite a major anti-nuclear campaign.

On 21 May 2017 there was a referendum on the government's Energy Strategy 2050, which was approved by a 58% majority, with voter turnout of 42%. It includes the provision for a gradual withdrawal from nuclear power and a greater reliance on hydro and intermittent renewables. Hydro currently supplies about 60% of the country's electricity, and generation from solar, wind, biomass and geothermal sources is to increase from 5 TWh in 2017 to at least 11.4 TWh by 2035. No construction licences will be issued for new nuclear power reactors and no "basic changes" to existing nuclear power plants will be permitted. The country's existing reactors will be allowed to remain in operation as long as ENSI considers them safe.

In August 2024, the Federal Council announced it would seek to lift the construction ban, citing climate targets, rising electricity demand, and geopolitical uncertainties. Energy Minister Albert Rösti stated the government was "responsible for leaving the door open to all possible technologies." The Federal Department of Energy submitted an amendment to the Nuclear Energy Act for consultation from December 2024 to April 2025. Following the consultation, in which the Swiss People's Party and business groups backed the proposal while the Social Democrats, Greens and a majority of cantons opposed it, the Federal Council adopted a dispatch to parliament in August 2025. A decision is expected by August 2026. The Greens have announced it would launch a referendum if the proposal passes.

Fuel cycle

Uranium is procured on world markets, enrichment is provided by a variety of contractors, and fuel fabrication is similarly diverse.

Radioactive waste management

Radioactive waste is mostly handled by Zwilag, a company owned by the four Swiss nuclear utilities. Its central interim dry cask storage facility for high-level waste (HLW), ZZL (Zentrales Zwischenlager), commenced operation in 2001 at Würenlingen. This is adjacent to the Paul Scherrer Institute, near NOK's Beznau nuclear power plant, and not far from two others. The Zwilag site also has facilities for incineration (in a high temperature plasma oven), conditioning and storage of low- and intermediate-level radioactive waste (LILW).

There has been no firm national policy regarding reprocessing or direct disposal of used fuel. However, until 2003 utilities were sending it for reprocessing so as to utilize the separated plutonium in mixed oxide (MOX) fuel. Reprocessing had been undertaken by Areva, at La Hague in France, and by BNFL at Sellafield in the UK under contract to individual power plant operators. Switzerland remained responsible for the separated high-level waste on return. About 1000 tonnes of used fuel was sent abroad for reprocessing, but the 2005 Nuclear Energy Act halted this for ten years from mid-2006, and the Energy Strategy 2050 extended the ban indefinitely. Used fuel is now retained at the reactors or sent to Zwilag ZZL for interim above-ground storage, being managed as high-level waste.

The Gösgen plant has limited pool capacity for used fuel storage so will operate an on-site independent fuel storage facility which allows cooling before used fuel is sent to Zwilag ZZL.

Repository plans for HLW

In 1972 a national co-operative for disposal of radioactive waste (NAGRA) was set up, involving power plant operators and the federal government. Since 1984 it has operated an underground research laboratory at Grimsel for HLW disposal.

NAGRA submitted a demonstration of feasibility of disposal report (Entsorgungsnachweis) to the Swiss government in 2002. The report showed that used fuel elements, separated high-level waste and long-lived intermediate-level waste could be safely disposed of in Switzerland. In June 2006, the Federal Council concluded that the legally required demonstration of disposal feasibility for all these had been successfully provided. Meanwhile the 2005 Nuclear Energy Act required the waste management and disposal program to proceed and be reviewed by the federal authorities. Identification of site options for disposal is proceeding under this Act and the Spatial Planning Act with regional participation, and following federal approval the actual site selection in three stages of the Sectoral Plan for Deep Geological Repositories is following.

In 2012 the Swiss Federal Office of Energy (SFOE) undertook a three-month consultation on NAGRA's 2008 repository plans considering six possible regions. NAGRA pursued investigation of these in 2014 in line with the requirements specified by ENSI, and two were short-listed in January 2015: in Jura Ost and Zurich Nordost. Each could accommodate both HLW and LILW repositories. Four sites of the original six were in reserve until December 2016 when Nördlich Lägern between the two others was added to the shortlist by ENSI after NAGRA had revived consideration of it. NAGRA had said in 2015: "The siting regions were evaluated and compared in a stepwise process, taking into consideration only scientific and technical criteria; societal and political aspects are not relevant in this respect." NAGRA began test drilling in Bulach in northern Lagern in April 2019. This was followed by drilling at Trüllikon in August 2019 at the Zurich Northeast site. The opalinus clay host rock in which the repository would be built lies about 900m deep.

The third phase of the process commissioned by the Federal Council in November 2018 will focus on these three regions, following a detailed review by ENSI of the reports and analyses submitted by NAGRA. NAGRA submitted its general licence

application in November 2024 to the Swiss Federal Office of Energy. The application also covers a fuel encapsulation plant at the Zwilag facility in Würenlingen. A final decision by the government is expected by 2030, with the possibility of a referendum. NAGRA expects a repository for LILW to be operating by 2050 and one for HLW by 2060.

LILW repository plans

A proposal for a low- and intermediate-level waste repository at Wellenberg was blocked by a cantonal referendum in 1995. A federal working group reviewed the proposal and recommended in 2000 that it proceed, though modified to allow for retrieval. A further cantonal referendum blocked it in 2002. The revised Nuclear Energy Act removes the cantonal veto right, but requires a national referendum.

Low- and intermediate-level waste from the nuclear power plants is processed into a form suitable for disposal either at sites of origin or at ZZL Würenlingen. It is packaged into suitable containers and then stored in facilities at the power plants or at ZZL. Two smaller interim storage sites for this waste have been operating since 1993: the government's BZL associated with the Paul Scherrer Institute at Würenlingen, and Zwibez at Beznau, which also has a storage hall for dry cask storage of spent fuel and high-level waste.

In September 2022, following a 14-year site selection process, Nagra proposed Nördlich Lägern in northern Switzerland as the site of a deep geological repository. The repository for LLW/ILW is planned to be in operation by 2050, with the one for HLW planned to be operational ten years later. Swissnuclear has been discussing with local communities and cantons possible payments and compensation for hosting the site.

Funding

The total cost of radioactive waste management is estimated at CHF 18.6 billion, including a 50-year post-operational monitoring phase. Nuclear plant owners pay into a national waste disposal fund created in 2002. A decommissioning fund was established in 1984 and power plant operators also pay annual contributions to this. The projected requirement is CHF 3.7 billion plus CHF 1.7 billion for post-operational preparation. At the end of 2017 the accumulated capital in both funds was CHF 7.7 billion, with CHF 5.8 billion having already been paid for waste management.

Both programmes are funded under the Nuclear Energy Act by a levy of about CHF 1 cent/kWh on nuclear power production.

Decommissioning

BKW is proceeding with decommissioning of the Mühleberg BWR following its shutdown in December 2019. Its operating licence was replaced with a decommissioning order issued by the Federal Department of Environment, Transport, Energy and Communication in September 2020. The work, costing about CHF 800 million, is expected to be completed in 2031, allowing relinquishment of the site by 2034. A further CHF 1.3 billion is budgeted for waste disposal in about 2040. BKW already has CHF 930 million in its decommissioning fund, and has made provision for more. All 418 fuel assemblies were transferred to the Würenlingen storage facility by September 2023, some 16 months ahead of schedule. About 4000 tonnes of demolition material too contaminated for cleaning and recycling will follow. The cutting and packaging of activated control rods was completed in October 2024.

Earlier, in March 2015 Alpiq set up a company, Swiss Decommissioning, to offer integrated solutions for post-operation and dismantling of nuclear installations, as well as for radiation protection and decontamination.

In September 2017, BKW acquired German radiation protection services company Dienstleistungen für Nukleartechnik GmbH. The German company has been providing radiation protection services to the Mühleberg plant since 2009.

Regulation, safety and non-proliferation

The main legislation governing nuclear energy is the 1959 Atomic Energy Act. It was updated in 1978 and 2003 (coming into force in 2005).

The Swiss Federal Nuclear Safety Inspectorate (ENSI, IFSN in French) is the national regulatory body with responsibility for the nuclear safety and security of Swiss nuclear facilities. It took over from the Swiss Federal Nuclear Safety Inspectorate (HSK) in 2009, but is an independent body rather than being under the Swiss Federal Office of Energy. It monitors and regulates both safety and radiological protection in nuclear installations and waste facilities. HSK was set up under the Federal Energy Office in 1982, and from 2003 there were legislative moves to make it independent, culminating in a 2007 Act setting up ENSI.

Civil liability for nuclear damage is covered by the 1983 and 2008 Nuclear Energy Liability Acts. Operators have unlimited liability, and they need to maintain insurance coverage. In March 2015 the minimum coverage was raised from CHF 1 billion to €1.2 billion. In 2009 Switzerland ratified the OECD Paris and Brussels conventions.

Non-proliferation

Switzerland is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1978. It is member of the Nuclear Suppliers' Group but not of Euratom. In 2000 it signed the Additional Protocol in relation to its safeguards agreements with the IAEA.

Public opinion

In November 2023, a poll by Demoscope on behalf of the Swiss Nuclear Forum found that 54% of respondents believed Switzerland should continue to use nuclear energy alongside renewables, up from 44% in February 2022. A September 2024 survey of some 19,500 people by the Leewas Institute showed majority support for the government's plan to lift the construction ban on new nuclear plants.

Notes & references

References

ENSI, NAGRA & Zwilag websites